

SQL & Pandas - Joins Exercise

Topic: Choosing the Correct Join

Instructions

- Use **Pandas** only.
 - **Do not modify** the original datasets.
 - Display the output after every question.
 - Decide the correct join based on the business scenario.
-

Dataset 1: Customers

```
import pandas as pd

customers = pd.DataFrame({
    "Customer_ID": ["C001", "C002", "C003", "C004", "C005", "C006"],
    "Customer_Name": ["Abel", "Sara", "John", "Helen", "Samuel",
"David"],
    "City": ["Addis Ababa", "Adama", "Jimma", "Bahir Dar", "Hawassa",
"Dire Dawa"]
})
```

Dataset 2: Orders

```
orders = pd.DataFrame({
    "Order_ID": [101,102,103,104,105,106,107],
    "Customer_ID": ["C001","C002","C001","C004","C007","C006","C006"],
    "Product":
    ["Laptop","Phone","Mouse","Keyboard","Tablet","Monitor","Printer"],
    "Amount": [1200,800,45,150,650,700,300]
})
```

Notes

- Customer **C005** has never placed an order.
- Order **105** belongs to **Customer C007**, who is not registered.

Exercise 1 — Customer Purchase Report

The sales manager wants a report that contains **only valid purchases made by registered customers**.

Create a report showing:

- Customer Name
- City
- Product
- Amount

Questions

1. How many records are returned?
2. Which order is excluded from the report?
3. Why was that order excluded?

Exercise 2 — Order Investigation

The finance department wants to review **every order**, even if customer information is unavailable.

Create a report showing:

- Order ID
- Customer ID
- Customer Name
- Product
- Amount

Questions

1. Which order has missing customer information?
 2. Why is the customer information missing?
 3. How many registered customers do not appear in this report?
-

Exercise 3 — Customer Engagement Report

The marketing team wants a list of **all registered customers**, including those who have never purchased anything.

Display:

- Customer Name
- City
- Product
- Amount

Questions

1. Which customer has never placed an order?
 2. What values appear for that customer's Product and Amount?
 3. Why do those values appear?
-

Exercise 4 — Data Quality Audit

The data engineering team wants to compare both datasets and identify any mismatches.

Create a report that includes every customer record and every order record.

Questions

1. How many rows are returned?
 2. Which registered customer has no orders?
 3. Which order belongs to a customer that is not registered?
 4. Why are both records important to keep during a data quality audit?
-

Exercise 5 — Customer Support Investigation

Customer **David** contacts customer support claiming he has placed several orders.

Generate a report showing:

- Customer Name
- Product
- Amount

Questions

1. How many orders has David placed?
2. What is his total spending?
3. Would this report include customers who have never placed orders? Explain your answer.